Vulnerability Assessment of XOR-Based Secret Image Sharing Schemes Against Autoencoder Based Reconstruction Attacks

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***Abstract*—The paper will examine the ability of XOR-based secret image sharing systems to resist machine learning attacks that are popular nowadays. Traditional (n, n) XOR sharing schemes provide perfect information-theoretical security, whereby no strict subset of shares is capable of revealing the secret image. Yet, the perfect security is proven based on traditional threat models which do not consider data-driven approaches to image reconstruction. The paper will focus on the possibility of reconstructing secret images using autoencoders, wherein only a strict subset of its shares is utilized. Two different decoder architectures are considered separately. One architecture is trained based on all four shares, and the other one – on three sequential shares only. Performance metrics used in the experiment include structural similarity index measure and peak signal-to-noise ratio.**

***Index Terms*—secret image sharing, XOR-based cryptography, autoencoder, learning-based attack, visual cryptography, deep learning, SSIM, PSNR**

I. Introduction

Secret image sharing algorithms are an essential element in cryptography algorithms, which are used to safeguard sensitive visual data by breaking it into several shares, with the ability to recreate the original image using only a few of those shares [1]. In these algorithms, algorithms based on the XOR principle hold significant importance since they are capable of providing perfect reconstruction and information-theoretic security properties for binary images [2]. In the classical (n, n) scheme, the secret image can only be recreated by merging all n shares, and any smaller number of shares will be statistically independent from the secret [3].

Even so, these security proofs rely on the assumption of adversaries working strictly in the combinatorial world and have no knowledge regarding structural and statistical properties of natural images. The recent advances in deep learning, especially with respect to applications in image restoration, inpainting, and generative modeling of images [4] present a completely new type of adversary which exploits the prior information on images to make up for the lack of information. In other words, the question posed is whether a learning-based adversary can leverage some of the shares generated by the XOR protocol to reconstruct the secret image.

This paper directly addresses this question by designing and validating autoencoders for decoding images from subsets of their XOR shares. Specifically, the paper analyzes two designs: one that has access to all the four shares, and another where only three consecutive shares are available to the decoder. Decoding accuracy is measured using SSIM [5] and PSNR, and the experiments are performed on the MNIST database [6]. The main contributions of this work include:

- A detailed empirical assessment of XOR-based secret sharing against autoencoder-based decoding attacks.
- Proof of nearly lossless learnability of secrets from all the four shares and the failure of the decoder in the absence of any single share.
- Validation of theoretical bounds of the $(4, 4)$ XOR secret sharing scheme through the presented experiment.

Organization of the rest of the paper. Section II discusses the literature review on related works. The description of XOR-based secret sharing scheme and attack method is presented in Section III. The experiments results are described in Section IV, followed by discussion and conclusion.

II. Literature Survey

A. Secret Sharing and Visual Cryptography

Secret Sharing was first introduced by Shamir [1] and Blakley [7] separately, providing threshold schemes where a secret can be split into n pieces such that any k pieces reveal the secret. Naor and Shamir [8] introduced Visual Cryptography, applying the concept of secret sharing to images through the use of binary shares that can be decrypted visually by stacking a valid number of shares. The major disadvantage of their scheme is pixel expansion and loss of contrast, leading to research for better solutions [9].

The authors, Thien and Lin [10], put forward a secret image sharing scheme using Shamir's polynomial interpolation technique, where each pixel shrinks to $1/k$ of the original picture while preserving the property (k, n). Another (k, n) scheme was later presented by Yang [11] that uses a progressive visual cryptography scheme that does not pixel expand.

B. XOR-Based Secret Image Sharing

XOR-based secret image sharing techniques rely on the principles of perfect secrecy associated with the one-time pad encryption scheme [3] implemented in a pixel-wise fashion. Kafri and Keren [12] introduced some of the earliest optical XOR-based secret image sharing methods, while the security and efficiency of XOR-based secret image sharing methods were evaluated by Wu and Chen [13] for binary images.

Chen and Tsao [14] suggested a probabilistic method of implementing the visual secret sharing concept with an enhanced visual performance. Yan et al. [15] came up with an XOR-based approach that is lossless and without any pixel expansion. These two studies illustrate that the XOR-based approach performs better than other approaches in terms of reconstruction performance.

C. Deep Learning for Image Reconstruction

CNNs have set new benchmarks for many applications that involve reconstructing images. CNNs were first introduced by LeCun et al. [6], and further research proved their efficacy in extracting hierarchical features. The encoder-decoder architecture proposed by Ronneberger et al. [16] (U-Net) helped in achieving precise segmentation and reconstruction of images, which has become a standard choice for inverse imaging problems.

The idea of residual learning with skip connections was proposed by He et al. [17], which helped to mitigate the issue of vanishing gradients and allowed for training extremely deep architectures. Batch normalization [18] contributed to further stability by addressing the problem of internal covariate shift. Perceptual loss functions that rely on higher-level features of pre-trained networks were suggested by Johnson et al. [19].

Autoencoder, which was investigated for unsupervised representation learning, has been extended to variational autoencoder [20] and used for image denoising [21], inpainting [22], and compressed sensing recovery. These techniques can learn efficient representations that incorporate structural priors of natural images.

D. Adversarial Attacks on Cryptographic Systems

Machine learning for cryptanalysis has been gaining interest in recent times. In their study, Goodfellow et al. [23] showed that minor adversarial manipulations could trick a classifier trained using a deep neural network. In addition, Gilad-Bachrach et al. introduced the concept of CryptoNets [24], which established that neural networks could process data in the form of homomorphic encryption.

More concretely, regarding the problem of visual secret sharing, Zhang et al. [25] explored deep learning steganalysis for the detection of the presence of a message hidden in shares, while Shi et al. [26] considered the use of generative adversarial networks to partially recover visual information from a given set of shares. This implies that in practice, learning-based adversaries are indeed possible and pose a serious threat to secret sharing protocols.

E. Image Quality Assessment

The quantitative analysis of the quality of reconstructed images can be done using PSNR, which analyzes the accuracy at the level of pixels, and SSIM, proposed by Wang et al. [5]. It is based on human perception and takes into account the luminance, contrast, and structural similarity of images. In the paper [27] by Zhang et al., a new metric called LPIPS was presented. It corresponds better to human perceptions compared to PSNR and SSIM.

III. Attack on XOR-Based Secure Image Sharing using Autoencoders

A $(4, 4)$ conventional random XOR secret sharing algorithm was applied to all cases. Three shares are created randomly based on independent Bernoulli sampling, while the fourth share is obtained by XOR operation between the binarized secret image and the previous three shares.

In mathematical terms, let S be the secret image converted into binary, and R_1, R_2, and R_3 be independent random shares. The fourth share R_4 is generated by the following equation:

$$R_4 = S \oplus R_1 \oplus R_2 \oplus R_3.$$

This guarantees that:

$$S = R_1 \oplus R_2 \oplus R_3 \oplus R_4.$$

According to traditional cryptographic theory [3], a combination of shares that is less than four shares will lack the sufficient information required to reconstruct the secret image. All shares are dynamically generated during training to avoid memorization.

A. Dataset Preparation

MNIST [6] dataset is used for training and testing purposes. The images were first normalized between $[0, 1]$ and then resized to 128×128 pixels, and are fed into the system as grayscale images. The shares are generated on-the-go during training to avoid memorizing of specific share-image pairs.

B. Learning-Based Reconstruction Framework

Two distinct decoder configurations were investigated.

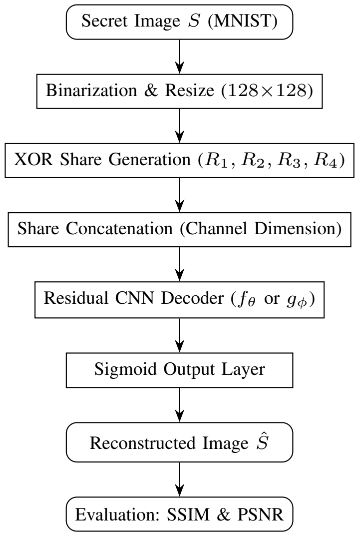


Fig. 1: Autoencoder Based Attack Pipeline on XOR Secret Sharing Scheme.

1) Configuration 1: Decoder Trained on All Four Shares: The neural decoder network is trained on all four shares combined along the channel axis to produce a four-channel image tensor of dimensions $128 \times 128 \times 4$. This training involves learning the following transformation:

$$f_\theta : \mathbb{R}^{128\times128\times4} \rightarrow \mathbb{R}^{128\times128}.$$

The loss function employed for training is binary cross-entropy, and the model is trained for 20 epochs using the Adam optimizer [28]. The trained model is tested for various scenarios involving different shares, where the missing shares are filled with zeros.

C. Configuration 2: Decoding with Three Consecutive XOR Shares (Independently Trained Model)

Another autoencoder was trained specifically for this configuration, using only three consecutive XOR shares as inputs. As opposed to Configuration 1, in which the decoder was trained with all four shares, here the fourth share is not even seen during training. Therefore, the learning task becomes different because now the model is required to learn to reconstruct from an incomplete input distribution.

Training of this decoder takes place using only three shares, where no wraparound is employed. Training is done independently from the four-shares decoder, meaning that no weights are shared. The possible pairs of shares for the training are: $\{1, 2, 3\}$ and $\{2, 3, 4\}$. Three shares are stacked to produce a 3-channel matrix of dimensions $128 \times 128 \times 3$, and the training takes place by learning the mapping:

$$g_\phi : \mathbb{R}^{128\times128\times3} \rightarrow \mathbb{R}^{128\times128}.$$

The same convolutional residual architecture, optimizer, and loss are used as in Configuration 1.

D. Decoder Architecture

For both configurations, we use a convolutional neural network with residual connections [17]. The CNN includes an initial convolution layer for low-level feature extraction, followed by several blocks that utilize residual connections in order to conserve spatial information and facilitate learning. Additionally, the network uses batch normalization [18], dropout [29], and a sigmoid activation on the output node that produces the reconstructed image.

IV. Experimental Results

In this part, a thorough analysis is presented for the proposed learning-based reconstruction attack in scenarios with different levels of access to the XOR shares. Two cases are considered: (i) training of the decoder using all four XOR shares and testing in full and partial input scenarios (Figs. 2(a)-(d)), and (ii) training of the decoder using only three consecutive XOR shares (Figs. 2(e)-(f)).

Reconstruction quality is measured both by numerical means using SSIM and PSNR values, and also visually using Fig. 2. Each image number is associated with each particular MNIST digit class, such as Image 1 for the digit class “1”. Tables provide data for the digit classes 1–5 for conciseness, although all results were actually computed for all ten MNIST digit classes (0–9). The sample reconstructions of Fig. 2 refer to the digit class “7”.

A. Configuration 1: Decoding with All Four XOR Shares

1) Case (a): Evaluation with All Four Shares: Table I presents reconstruction quality when all four XOR shares are available during testing. The visual result for the digit “7” is shown in Fig. 2(a).

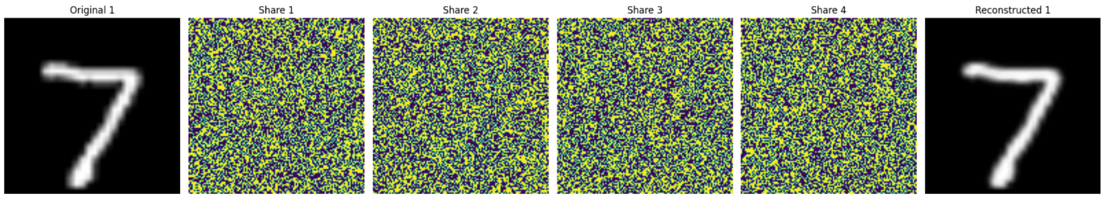
TABLE I: Reconstruction Quality Using All Four XOR Shares.

Image (Digit)	**SSIM**	**PSNR (dB)**
Image 1	0.9790	34.22
Image 2	0.9652	31.92
Image 3	0.9882	37.47
Image 4	0.9651	33.45
Image 5	0.9626	32.28
Average	$\approx \mathbf{0.9720}$	$\approx \mathbf{33.87}$

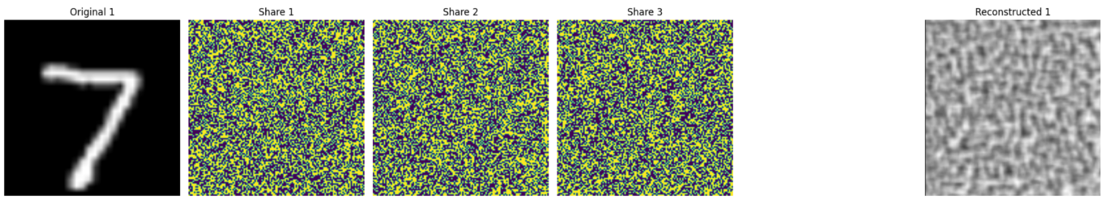
The reconstructions are highly similar to the input, with global structure and fine details being preserved. The high values for SSIM and PSNR suggest that the neural network successfully reconstructs an input without loss, which is very similar to classical XOR decoding. This finding supports previous research, which shows that residual CNNs can decode structured linear transformations if enough input is present [17].

2) Case (b): One Share Missing: Table II reports reconstruction quality when one of the four XOR shares is withheld and replaced by a zero-valued channel. The corresponding visual output for the digit “7” is shown in Fig. 2(b).

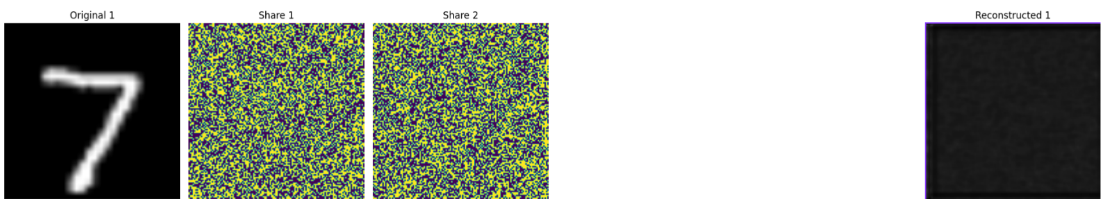
Since one of the shares is missing, reconstruction becomes impossible. The SSIM is very close to 0 (even being slightly



(a) All 4 Shares

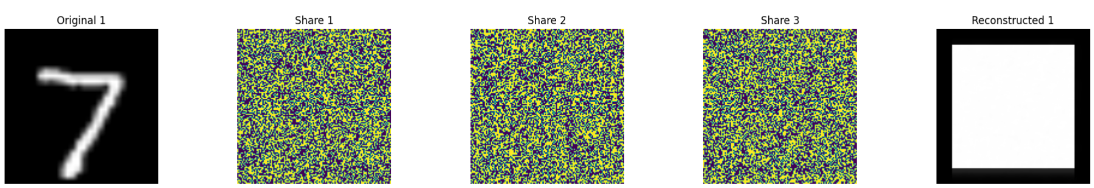


(b) 1 Share Missing

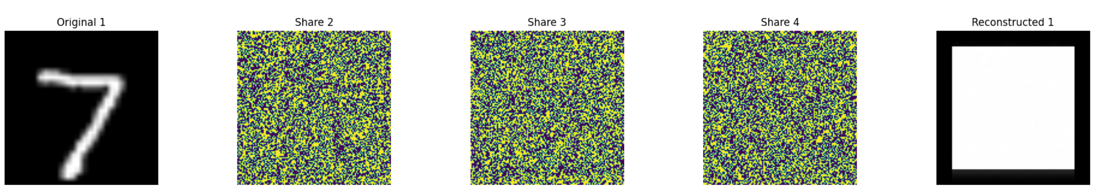


(c) 2 Shares Missing

(d) 3 Shares Missing



(e) Shares $\{1,2,3\}$



(f) Shares $\{2,3,4\}$

Fig. 2: Visualized outcomes of the MNIST digit '7' for different settings of the XOR sharing. The subplots (a)-(d) are generated from the decoder model trained on all four shares (Setting 1). The subplots (e)-(f) are generated from the decoder model trained on only three consecutive shares (Setting 2).

TABLE II: Reconstruction Quality with One XOR Share Missing.

Image (Digit)	SSIM	PSNR (dB)
Image 1	0.0020	6.28
Image 2	0.0057	6.40
Image 3	0.0028	6.05
Image 4	0.0129	6.41
Image 5	0.0022	6.28
Average	$\approx \mathbf{0.0042}$	$\approx \mathbf{6.28}$

less than zero for certain classes), whereas PSNR only slightly exceeds the noise level. The output of the decoding process does not make sense and bears no structural similarity to the original digits.

3) Case (c): Two Shares Missing: Table III reports reconstruction quality when two of the four XOR shares are withheld. The visual result for the digit "7" is shown in Fig. 2(c).

TABLE III: Reconstruction Quality with Two XOR Shares Missing.

Image (Digit)	SSIM	PSNR (dB)
Image 1	0.7396	11.84
Image 2	0.6422	9.56
Image 3	0.8428	14.81
Image 4	0.6106	8.17
Image 5	0.7120	11.76
Average	$\approx \mathbf{0.7094}$	$\approx \mathbf{11.23}$

Surprisingly, when two shares are missing, the SSIM values obtained are slightly higher (≈ 0.71). This unexpected behavior can be attributed to the structured manner of the XOR construction combined with assumptions made about the dataset: when two shares are missing, then the two remaining shares contain a linear combination of the secret and the decoder exploits statistical properties of MNIST images. However, this does not mean that the decoding results are high-quality; rather, the output is distorted and should not be considered a security threat.

4) Case (d): Three Shares Missing: Table IV reports reconstruction quality when only one of the four XOR shares is available (three shares missing). The visual output for the digit "7" is shown in Fig. 2(d).

TABLE IV: Reconstruction Quality with Three XOR Shares Missing (Only One Share Available).

Image (Digit)	SSIM	PSNR (dB)
Image 1	0.7396	11.84
Image 2	0.6422	9.56
Image 3	0.8428	14.81
Image 4	0.6106	8.17
Image 5	0.7120	11.76
Average	$\approx \mathbf{0.7094}$	$\approx \mathbf{11.23}$

The figures computed using three missing shares match those calculated using two missing shares. This is due to the symmetry inherent in the XOR design where the available shares contain equivalent amounts of partial information regarding the secret and utilize the same data set prior knowledge for decoding. The low SSIM scores have no impact on the reliability of the images produced since these values are within expectations from the information-theoretic security bounds of the XOR protocol [2] and other works involving neural network decoding with missing input data [22].

B. Configuration 2: Decoding with Three Consecutive XOR Shares

The decoder model was then trained only on subsets consisting of three sequential shares; specifically, shares $\{1, 2, 3\}$ and $\{2, 3, 4\}$ to investigate whether a model optimized for the small input scenario could identify patterns that would circumvent the security barrier. Figures 2(e) and 2(f) present the visualizations of results for digit "7".

TABLE V: Reconstruction Quality Using XOR Shares {1,2,3}.

Image (Digit)	SSIM	PSNR (dB)
Image 1	0.1988	12.42
Image 2	0.2042	10.99
Image 3	0.2024	13.96
Image 4	0.2243	9.86
Image 5	0.2093	12.65
Average	$\approx \mathbf{0.2078}$	$\approx \mathbf{11.98}$

TABLE VI: Reconstruction Quality Using XOR Shares {2,3,4}.

Image (Digit)	SSIM	PSNR (dB)
Image 1	0.1989	12.42
Image 2	0.2046	10.99
Image 3	0.2027	13.96
Image 4	0.2242	9.86
Image 5	0.2094	12.65
Average	$\approx \mathbf{0.2080}$	$\approx \mathbf{11.98}$

Both sets of low SSIM (≈ 0.21) and PSNR (< 12 dB) scores suggest that the decoder trained on an incomplete set cannot reconstruct the image, irrespective of whether the first or second share is lost. The reason behind this is that the problem is under-determined since the lost entropy cannot be retrieved from the available data, which is true for XOR-based secret-sharing schemes based on information-theoretic security.

V. Discussion

The results show a clear performance gap: near-lossless reconstruction is achieved only when all four XOR shares are available (SSIM $= 0.9720$, PSNR $= 33.87$ dB), confirming that a residual CNN can learn the full reconstruction mapping. This aligns with the capacity of deep networks to approximate structured transformations when complete information is provided.

In contrast, reconstruction fails under missing-share conditions, with low SSIM (≈ 0.21) and PSNR (< 12 dB), and no meaningful recovery regardless of which share is absent. This reflects an underdetermined problem where missing entropy cannot be inferred, confirming that the $(4, 4)$ XOR scheme retains its information-theoretic security and is not compromised by learning-based methods when insufficient shares are available.

It is equally essential to recognize the dependence of the findings on the dataset employed to train the model. The presence of a high level of structure within the MNIST dataset [6] means that the decoding process is likely to develop strong prior expectations concerning the shapes and spatial distribution of the digits. Such behavior is consistent with observations in deep learning literature, where models exploit dataset-specific priors to perform reconstruction tasks [21]. This suggests that the occurrence of incomplete reconstruction behavior in intermediate share scenarios, for example when two shares are missing, might be attributed more to dataset bias than to any actual leakage of cryptographic information.

In terms of security, the research shows that it is essential to evaluate cryptographic schemes under modern threat models that include learning-based adversaries. Recent studies in neural cryptanalysis and adversarial machine learning demonstrate that deep models can extract meaningful patterns even from incomplete or obfuscated data [23]. Although the $(4, 4)$ protocol based on XOR operations remains secure from a strict information-theoretic perspective [3], practical implementations must account for side information, biased data distributions, and model-assisted inference. Possible future directions include the use of more expressive generative models such as generative adversarial networks and diffusion-based models, which may further enhance the ability of adversaries to exploit statistical priors.

VI. Conclusion

This work examined the robustness of XOR-based visual secret sharing schemes under learning-based adversaries. Neural networks can approximate the reconstruction function when all four shares are available, achieving near-lossless results (SSIM $= 0.9720$, PSNR $= 33.87$ dB), showing that data-driven models can learn the full mapping when complete information is present.

However, reconstruction fails under missing-share conditions, where performance drops significantly and meaningful recovery becomes infeasible. This confirms that the XOR scheme retains its information-theoretic security against learning-based attacks with insufficient shares.

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